

AERIS Expert Review Packet

Anomaly

Source / metric	tceq / Nitrogen dioxide (no2)
Observed / expected baseline	18.8 / 6.01223 ppb
Baseline method	mean of the preceding 7 days of this series (Z-score detector)
Time	2026-07-15 21:00 UTC
Location	29.6863, -95.2947
Severity	severe
Detectors	zscore, stl, isolation_forest

Stored evidence summary

The three tables in this section are look-up tables. You never have to read them through: they are here for the moment a claim quotes a number and you want to check it.

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
asos	Relative humidity (humidity)	percent	9 / 908	47.55 to 100; mean 84.0193	69.97 at 2026-07-15 20:55Z; 5 min before event
asos	Air temperature (temperature)	degC	9 / 908	22 to 34.4444; mean 26.5318	28 at 2026-07-15 20:55Z; 5 min before event
asos	Wind direction (wind_direction)	degree	9 / 1201	0 to 360; mean 106.486	0 at 2026-07-15 20:55Z; 5 min before event
asos	Wind speed (wind_speed)	m/s	9 / 1208	0 to 10.2889; mean 2.8878	0 at 2026-07-15 20:55Z; 5 min before event
noaa_gfs	500-hPa geopotential height (gh_500)	m	11 / 132	5881.42 to 5937.36; mean 5904.38	5897.38 at 2026-07-15 18:00Z; 3 h before event
noaa_gfs	Planetary boundary layer height (pbl_height)	m	11 / 132	21.6294 to 1860.82; mean 366.522	113.722 at 2026-07-15 18:00Z; 3 h before event
noaa_gfs	Precipitable water (precipitable_water)	mm	11 / 132	42.269 to 63.2136; mean 48.6642	50.2845 at 2026-07-15 18:00Z; 3 h before event
noaa_gfs	Surface pressure (surface_pressure)	hPa	11 / 132	1011.55 to 1019.21; mean 1015.3	1018.07 at 2026-07-15 18:00Z; 3 h before event
noaa_gfs	850-hPa temperature (t_850)	degC	11 / 132	16.0656 to 19.7475; mean 18.2499	17.3174 at 2026-07-15 18:00Z; 3 h before event
noaa_gfs	10-m eastward wind (u_10m)	m/s	11 / 132	-4.73694 to 1.38902; mean -1.5836	-1.73098 at 2026-07-15 18:00Z; 3 h before event
noaa_gfs	10-m northward wind (v_10m)	m/s	11 / 132	-2.16541 to 5.42479; mean 1.4262	-1.30506 at 2026-07-15 18:00Z; 3 h before event
openaq	Ozone (ozone)	ppm	17 / 1147	-0.003 to 0.038; mean 0.0127	0.002 at 2026-07-15 21:00Z; at event time

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
openaq	PM10 (pm10)	ug/m3	2 / 143	14 to 74; mean 38.3217	34 at 2026-07-15 21:00Z; at event time
openaq	PM2.5 (pm25)	ug/m3	79 / 4546	0 to 32.7; mean 6.0478	4.2 at 2026-07-15 21:00Z; at event time
openweather	Cloud cover (cloud_cover)	percent	5 / 359	0 to 100; mean 93.6045	100 at 2026-07-15 21:01Z; 2 min after event
openweather	Relative humidity (humidity)	percent	5 / 359	49 to 97; mean 86.0251	76 at 2026-07-15 21:01Z; 2 min after event
openweather	Precipitation rate (precipitation)	mm/h	5 / 359	0 to 15.12; mean 0.2914	0 at 2026-07-15 21:01Z; 2 min after event
openweather	Surface pressure (pressure)	hPa	5 / 359	1016 to 1021; mean 1017.79	1018 at 2026-07-15 21:01Z; 2 min after event
openweather	Air temperature (temperature)	degC	5 / 359	22.54 to 34.2; mean 26.6957	27.56 at 2026-07-15 21:01Z; 2 min after event
openweather	Wind direction (wind_direction)	degree	5 / 359	0 to 355; mean 130.131	113 at 2026-07-15 21:01Z; 2 min after event
openweather	Wind speed (wind_speed)	m/s	5 / 359	0.34 to 5.81; mean 1.8992	0.45 at 2026-07-15 21:01Z; 2 min after event
purpleair	PM2.5 (pm25)	ug/m3	71 / 4861	0 to 45.7; mean 7.2635	5.4 at 2026-07-15 21:00Z; at event time
sentinel5p	Aerosol index granules (s5p_aer_ai_granule_available)	count	6 / 6	1 to 1; mean 1	1 at 2026-07-15 19:45Z; 1 h 15 min before event
sentinel5p	CH4 granules available (s5p_ch4_granule_available)	count	6 / 6	1 to 1; mean 1	1 at 2026-07-15 19:45Z; 1 h 15 min before event
sentinel5p	CO column (s5p_co_column)	mol/m^2	2 / 2	0.0224462 to 0.0224492; mean 0.0224	0.0224492 at 2026-07-16 18:58Z; 21 h 59 min after event
sentinel5p	CO granules available (s5p_co_granule_available)	count	6 / 6	1 to 1; mean 1	1 at 2026-07-15 19:45Z; 1 h 15 min before event
sentinel5p	HCHO column (s5p_hcho_column)	mol/m^2	1 / 1	0.000134449 to 0.000134449; mean 0.0001	0.000134449 at 2026-07-16 19:25Z; 22 h 25 min after event
sentinel5p	HCHO granules available (s5p_hcho_granule_available)	count	5 / 5	1 to 1; mean 1	1 at 2026-07-15 19:45Z; 1 h 15 min before event
sentinel5p	NO2 column (s5p_no2_column)	mol/m^2	1 / 1	3.22068e-05 to 3.22068e-05; mean 0	3.22068e-05 at 2026-07-16 19:25Z; 22 h 25 min after event
sentinel5p	NO2 granules available (s5p_no2_granule_available)	count	3 / 3	1 to 1; mean 1	1 at 2026-07-15 19:45Z; 1 h 15 min before event
sentinel5p	O3 granules available (s5p_o3_granule_available)	count	6 / 6	1 to 1; mean 1	1 at 2026-07-15 19:45Z; 1 h 15 min before event
sentinel5p	SO2 column (s5p_so2_column)	mol/m^2	1 / 1	-2.09205e-05 to -2.09205e-05; mean -0	-2.09205e-05 at 2026-07-16 19:25Z; 22 h 25 min after event
sentinel5p	SO2 granules available (s5p_so2_granule_available)	count	5 / 5	1 to 1; mean 1	1 at 2026-07-15 19:45Z; 1 h 15 min before event
tceq	Carbon monoxide (co)	ppm	3 / 172	0.1 to 0.9; mean 0.2372	0.2 at 2026-07-15 21:00Z; at event time
tceq	Nitrogen dioxide (no2)	ppb	12 / 757	-0.3 to 38.4; mean 7.5758	18.8 at 2026-07-15 21:00Z; at event time

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
tceq	Sulfur dioxide (so2)	ppb	3 / 190	-0.6 to 1.2; mean -0.2837	-0.4 at 2026-07-15 21:00Z; at event time

Six-hour averages

Each metric averaged in 6-hour blocks across the stored 72 hours, in UTC. These averages, and the per-site ones below, are what the models were shown.

Source	Metric	Values
asos	Relative humidity (humidity)	07-14 06Z 94.28, 12Z 96.12, 18Z 87.53, 07-15 00Z 88.3, 06Z 87.86, 12Z 81.17, 18Z 72.97, 07-16 00Z 84.8, 06Z 93.75, 12Z 75.57, 18Z 62.47, 07-17 00Z 81.28, 06Z 88.18
asos	Air temperature (temperature)	07-14 06Z 23.95, 12Z 23.41, 18Z 25.58, 07-15 00Z 25.43, 06Z 25.13, 12Z 25.01, 18Z 27.95, 07-16 00Z 26.59, 06Z 25.45, 12Z 29.81, 18Z 32.04, 07-17 00Z 28.1, 06Z 26.7
asos	Wind direction (wind_direction)	07-14 06Z 136, 12Z 87.06, 18Z 108.8, 07-15 00Z 75.7, 06Z 82.73, 12Z 127.2, 18Z 77.84, 07-16 00Z 79.2, 06Z 78.14, 12Z 141.8, 18Z 152.9, 07-17 00Z 146.3, 06Z 106.9
asos	Wind speed (wind_speed)	07-14 06Z 2.943, 12Z 2.724, 18Z 2.671, 07-15 00Z 2.705, 06Z 2.676, 12Z 2.567, 18Z 2.406, 07-16 00Z 2.109, 06Z 1.654, 12Z 3.705, 18Z 5.465, 07-17 00Z 3.435, 06Z 2.159
noaa_gfs	500-hPa geopotential height (gh_500)	07-14 12Z 5899, 18Z 5900, 07-15 00Z 5890, 06Z 5892, 12Z 5884, 18Z 5896, 07-16 00Z 5896, 06Z 5909, 12Z 5906, 18Z 5926, 07-17 00Z 5919, 06Z 5935
noaa_gfs	Planetary boundary layer height (pbl_height)	07-14 12Z 124.1, 18Z 253.7, 07-15 00Z 141.4, 06Z 153.5, 12Z 186.1, 18Z 142.3, 07-16 00Z 319, 06Z 127.9, 12Z 110.3, 18Z 1625, 07-17 00Z 860.3, 06Z 354.4
noaa_gfs	Precipitable water (precipitable_water)	07-14 12Z 58.42, 18Z 51.56, 07-15 00Z 50.38, 06Z 50.1, 12Z 47.73, 18Z 50.25, 07-16 00Z 48.02, 06Z 43.75, 12Z 44.93, 18Z 45.34, 07-17 00Z 46.17, 06Z 47.33
noaa_gfs	Surface pressure (surface_pressure)	07-14 12Z 1015, 18Z 1017, 07-15 00Z 1014, 06Z 1016, 12Z 1015, 18Z 1017, 07-16 00Z 1014, 06Z 1015, 12Z 1015, 18Z 1016, 07-17 00Z 1014, 06Z 1016
noaa_gfs	850-hPa temperature (t_850)	07-14 12Z 16.83, 18Z 17.35, 07-15 00Z 18.3, 06Z 17.53, 12Z 18.71, 18Z 17.37, 07-16 00Z 19.05, 06Z 19.22, 12Z 18.87, 18Z 17.82, 07-17 00Z 18.64, 06Z 19.3
noaa_gfs	10-m eastward wind (u_10m)	07-14 12Z -0.6575, 18Z -2.077, 07-15 00Z -1.727, 06Z -2.203, 12Z -2.582, 18Z -0.7246, 07-16 00Z -1.768, 06Z -1.734, 12Z -0.4817, 18Z -1.54, 07-17 00Z -2.477, 06Z -1.031
noaa_gfs	10-m northward wind (v_10m)	07-14 12Z 1.35, 18Z -0.5181, 07-15 00Z 0.3779, 06Z -0.7867, 12Z 0.9323, 18Z -0.3978, 07-16 00Z 1.357, 06Z 1.628, 12Z 1.857, 18Z 4.222, 07-17 00Z 4.468, 06Z 2.626
openaq	Ozone (ozone)	07-14 06Z 0.007667, 12Z 0.01045, 18Z 0.01501, 07-15 00Z 0.01197, 06Z 0.007833, 12Z 0.01287, 18Z 0.01966, 07-16 00Z 0.01489, 06Z 0.003919, 12Z 0.009939, 18Z 0.0219, 07-17 00Z 0.01324, 06Z 0.009698
openaq	PM10 (pm10)	07-14 06Z 28.83, 12Z 19.5, 18Z 21.75, 07-15 00Z 24.25, 06Z 28.42, 12Z 48.91, 18Z 33.92, 07-16 00Z 42.75, 06Z 51.58, 12Z 56.08, 18Z 52, 07-17 00Z 43.5, 06Z 40.38
openaq	PM2.5 (pm25)	07-14 06Z 4.574, 12Z 3.461, 18Z 4.584, 07-15 00Z 4.715, 06Z 4.704, 12Z 5.603, 18Z 5.996, 07-16 00Z 6.576, 06Z 10.04, 12Z 11.41, 18Z 6.704, 07-17 00Z 6.582, 06Z 5.584
openweather	Cloud cover (cloud_cover)	07-14 06Z 98.14, 12Z 99.71, 18Z 99.87, 07-15 00Z 99.97, 06Z 100, 12Z 100, 18Z 100, 07-16 00Z 99.03, 06Z 94.77, 12Z 99.67, 18Z 98.4, 07-17 00Z 44.37, 06Z 75.5
openweather	Relative humidity (humidity)	07-14 06Z 94.21, 12Z 95.19, 18Z 92.23, 07-15 00Z 90.53, 06Z 92.24, 12Z 86.68, 18Z 79.2, 07-16 00Z 83.77, 06Z 92.97, 12Z 83.33, 18Z 66.03, 07-17 00Z 79.4, 06Z 87.64
openweather	Precipitation rate (precipitation)	07-14 06Z 1.534, 12Z 1.84, 18Z 0.301, 07-15 00Z 0, 06Z 0, 12Z 0.5113, 18Z 0.041, 07-16 00Z 0, 06Z 0, 12Z 0, 18Z 0, 07-17 00Z 0, 06Z 0
openweather	Surface pressure (pressure)	07-14 06Z 1016, 12Z 1018, 18Z 1018, 07-15 00Z 1017, 06Z 1017, 12Z 1019, 18Z 1018, 07-16 00Z 1017, 06Z 1018, 12Z 1019, 18Z 1018, 07-17 00Z 1018, 06Z 1019
openweather	Air temperature (temperature)	07-14 06Z 24.57, 12Z 23.28, 18Z 25.11, 07-15 00Z 25.41, 06Z 24.94, 12Z 25.15, 18Z 27.01, 07-16 00Z 27.07, 06Z 25.41, 12Z 29.12, 18Z 33.16, 07-17 00Z 28.9, 06Z 27.14
openweather	Wind direction (wind_direction)	07-14 06Z 151.5, 12Z 164.6, 18Z 120.7, 07-15 00Z 71.7, 06Z 101.7, 12Z 139.1, 18Z 105.2, 07-16 00Z 124.6, 06Z 143, 12Z 154.9, 18Z 155.2, 07-17 00Z 136, 06Z 135.1
openweather	Wind speed (wind_speed)	07-14 06Z 1.441, 12Z 2.031, 18Z 1.482, 07-15 00Z 1.853, 06Z 1.493, 12Z 1.667, 18Z 2.515, 07-16 00Z 1.716, 06Z 1.639, 12Z 1.863, 18Z 2.737, 07-17 00Z 2.168, 06Z 1.751

Source	Metric	Values
purpleair	PM2.5 (pm25)	07-14 06Z 5.145, 12Z 3.648, 18Z 5.262, 07-15 00Z 5.223, 06Z 5.152, 12Z 6.768, 18Z 6.637, 07-16 00Z 8.205, 06Z 12.28, 12Z 13.29, 18Z 8.049, 07-17 00Z 7.99, 06Z 7.815
sentinel5p	Aerosol index granules (s5p_aer_ai_granule_available)	07-14 18Z 1, 07-15 18Z 1, 07-16 18Z 1
sentinel5p	CH4 granules available (s5p_ch4_granule_available)	07-14 18Z 1, 07-15 18Z 1, 07-16 18Z 1
sentinel5p	CO granules available (s5p_co_granule_available)	07-14 18Z 1, 07-15 18Z 1, 07-16 18Z 1
sentinel5p	HCHO granules available (s5p_hcho_granule_available)	07-14 18Z 1, 07-15 18Z 1, 07-16 18Z 1
sentinel5p	NO2 granules available (s5p_no2_granule_available)	07-14 18Z 1, 07-15 18Z 1, 07-16 18Z 1
sentinel5p	O3 granules available (s5p_o3_granule_available)	07-14 18Z 1, 07-15 18Z 1, 07-16 18Z 1
sentinel5p	SO2 granules available (s5p_so2_granule_available)	07-14 18Z 1, 07-15 18Z 1, 07-16 18Z 1
tceq	Carbon monoxide (co)	07-14 06Z 0.2111, 12Z 0.2, 18Z 0.2667, 07-15 00Z 0.17, 06Z 0.1444, 12Z 0.3, 18Z 0.2222, 07-16 00Z 0.34, 06Z 0.2611, 12Z 0.3214, 18Z 0.2083, 07-17 00Z 0.2333, 06Z 0.1625
tceq	Nitrogen dioxide (no2)	07-14 06Z 6.825, 12Z 9.866, 18Z 8.889, 07-15 00Z 7.682, 06Z 7.693, 12Z 9.772, 18Z 10.24, 07-16 00Z 10.26, 06Z 7.385, 12Z 6.061, 18Z 3.566, 07-17 00Z 3.7, 06Z 4.67
tceq	Sulfur dioxide (so2)	07-14 06Z -0.3778, 12Z -0.3111, 18Z -0.2667, 07-15 00Z -0.2778, 06Z -0.3187, 12Z -0.2611, 18Z -0.1167, 07-16 00Z -0.1222, 06Z -0.35, 12Z -0.3222, 18Z -0.3056, 07-17 00Z -0.3, 06Z -0.3583

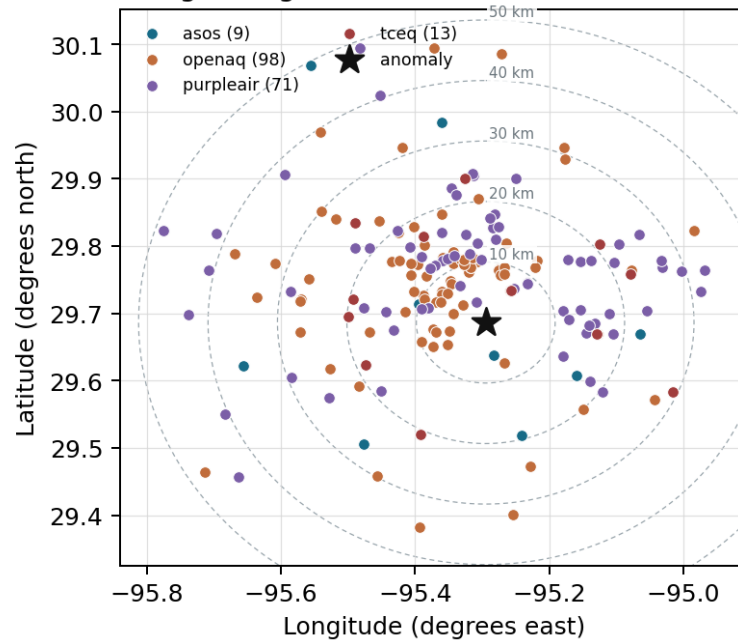
Per-site averages

Each metric averaged at each site, with that site's distance from the event in brackets.

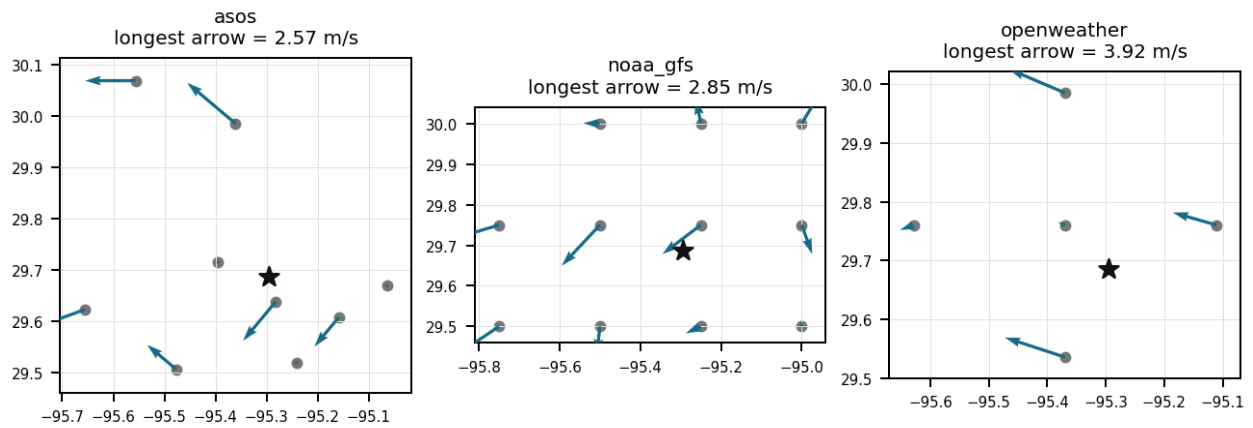
Source	Metric	Values
asos	Relative humidity (humidity)	HOU (5.555 km) 82.22, MCJ (10.163 km) 83.02, EFD (15.811 km) 84.78, LVJ (19.307 km) 82.42, T41 (22.362 km) 83.55, AXH (26.678 km) 81.79, IAH (33.753 km) 87.24, SGR (35.675 km) 86.67, +1 more
asos	Air temperature (temperature)	HOU (5.555 km) 26.78, MCJ (10.163 km) 26.77, EFD (15.811 km) 26.73, LVJ (19.307 km) 26.47, T41 (22.362 km) 27.01, AXH (26.678 km) 26.36, IAH (33.753 km) 26.17, SGR (35.675 km) 26.32, +1 more
asos	Wind direction (wind_direction)	HOU (5.555 km) 121.6, MCJ (10.163 km) 132.3, EFD (15.811 km) 88.1, LVJ (19.307 km) 92.29, T41 (22.362 km) 123.5, AXH (26.678 km) 78.46, IAH (33.753 km) 109, SGR (35.675 km) 116.2, +1 more
asos	Wind speed (wind_speed)	HOU (5.555 km) 3.322, MCJ (10.163 km) 4.329, EFD (15.811 km) 2.825, LVJ (19.307 km) 2.326, T41 (22.362 km) 3.561, AXH (26.678 km) 1.432, IAH (33.753 km) 3.351, SGR (35.675 km) 2.958, +1 more
noaa_gfs	500-hPa geopotential height (gh_500)	gfs:29.75,-95.25 (8.297 km) 5905, gfs:29.75,-95.50 (21.05 km) 5904, gfs:29.50,-95.25 (21.162 km) 5905, gfs:29.50,-95.50 (28.689 km) 5905, gfs:29.75,-95.00 (29.33 km) 5905, gfs:30.00,-95.25 (35.148 km) 5904, gfs:29.50,-95.00 (35.231 km) 5906, gfs:30.00,-95.50 (40.109 km) 5903, +3 more
noaa_gfs	Planetary boundary layer height (pbl_height)	gfs:29.75,-95.25 (8.297 km) 371.8, gfs:29.75,-95.50 (21.05 km) 458.6, gfs:29.50,-95.25 (21.162 km) 285.9, gfs:29.50,-95.50 (28.689 km) 322.1, gfs:29.75,-95.00 (29.33 km) 330.1, gfs:30.00,-95.25 (35.148 km) 410.9, gfs:29.50,-95.00 (35.231 km) 400.4, gfs:30.00,-95.50 (40.109 km) 483, +3 more
noaa_gfs	Precipitable water (precipitable_water)	gfs:29.75,-95.25 (8.297 km) 48.91, gfs:29.75,-95.50 (21.05 km) 48.81, gfs:29.50,-95.25 (21.162 km) 48.61, gfs:29.50,-95.50 (28.689 km) 48.92, gfs:29.75,-95.00 (29.33 km) 48.64, gfs:30.00,-95.25 (35.148 km) 48.82, gfs:29.50,-95.00 (35.231 km) 48.51, gfs:30.00,-95.50 (40.109 km) 48.43, +3 more

Source	Metric	Values
noaa_gfs	Surface pressure (surface_pressure)	gfs:29.75,-95.25 (8.297 km) 1016, gfs:29.75,-95.50 (21.05 km) 1015, gfs:29.50,-95.25 (21.162 km) 1016, gfs:29.50,-95.50 (28.689 km) 1015, gfs:29.75,-95.00 (29.33 km) 1017, gfs:30.00,-95.25 (35.148 km) 1015, gfs:29.50,-95.00 (35.231 km) 1017, gfs:30.00,-95.50 (40.109 km) 1013, +3 more
noaa_gfs	850-hPa temperature (t_850)	gfs:29.75,-95.25 (8.297 km) 18.19, gfs:29.75,-95.50 (21.05 km) 18.19, gfs:29.50,-95.25 (21.162 km) 18.44, gfs:29.50,-95.50 (28.689 km) 18.39, gfs:29.75,-95.00 (29.33 km) 18.27, gfs:30.00,-95.25 (35.148 km) 18.15, gfs:29.50,-95.00 (35.231 km) 18.39, gfs:30.00,-95.50 (40.109 km) 18.01, +3 more
noaa_gfs	10-m eastward wind (u_10m)	gfs:29.75,-95.25 (8.297 km) -1.376, gfs:29.75,-95.50 (21.05 km) -1.477, gfs:29.50,-95.25 (21.162 km) -1.802, gfs:29.50,-95.50 (28.689 km) -1.851, gfs:29.75,-95.00 (29.33 km) -1.35, gfs:30.00,-95.25 (35.148 km) -1.434, gfs:29.50,-95.00 (35.231 km) -1.717, gfs:30.00,-95.50 (40.109 km) -1.421, +3 more
noaa_gfs	10-m northward wind (v_10m)	gfs:29.75,-95.25 (8.297 km) 1.055, gfs:29.75,-95.50 (21.05 km) 1.178, gfs:29.50,-95.25 (21.162 km) 1.652, gfs:29.50,-95.50 (28.689 km) 1.462, gfs:29.75,-95.00 (29.33 km) 1.358, gfs:30.00,-95.25 (35.148 km) 1.055, gfs:29.50,-95.00 (35.231 km) 2.409, gfs:30.00,-95.50 (40.109 km) 1.152, +3 more
openaq	Ozone (ozone)	325 (0.01 km) 0.003354, 3378 (6.399 km) 0.006866, 320 (7.255 km) 0.01179, 2039 (11.575 km) 0.01051, 1319333 (13.71 km) 0.01355, 2046 (15.8 km) 0.01399, 332 (16.16 km) 0.01838, 312 (18.679 km) 0.01373, +9 more
openaq	PM10 (pm10)	15852419 (9.465 km) 36.47, 271 (16.16 km) 40.2
openaq	PM2.5 (pm25)	14140751 (0.011 km) 6.072, 15990898 (4.432 km) 4.596, 16564662 (4.852 km) 4.766, 13418566 (5.486 km) 5.632, 8967123 (6.375 km) 8.641, 8967479 (6.375 km) 7.922, 15539682 (6.38 km) 7.21, 15472101 (6.665 km) 4.396, +71 more
openweather	Cloud cover (cloud_cover)	grid:center (10.975 km) 93.38, grid:south (18.24 km) 92.57, grid:east (19.553 km) 96.74, grid:west (33.267 km) 91.06, grid:north (33.994 km) 94.28
openweather	Relative humidity (humidity)	grid:center (10.975 km) 85.41, grid:south (18.24 km) 83.28, grid:east (19.553 km) 87.44, grid:west (33.267 km) 86.61, grid:north (33.994 km) 87.38
openweather	Precipitation rate (precipitation)	grid:center (10.975 km) 0.1166, grid:south (18.24 km) 0.3574, grid:east (19.553 km) 0.2251, grid:west (33.267 km) 0.3876, grid:north (33.994 km) 0.3681
openweather	Surface pressure (pressure)	grid:center (10.975 km) 1018, grid:south (18.24 km) 1018, grid:east (19.553 km) 1018, grid:west (33.267 km) 1018, grid:north (33.994 km) 1018
openweather	Air temperature (temperature)	grid:center (10.975 km) 26.98, grid:south (18.24 km) 26.76, grid:east (19.553 km) 26.67, grid:west (33.267 km) 26.5, grid:north (33.994 km) 26.57
openweather	Wind direction (wind_direction)	grid:center (10.975 km) 127.8, grid:south (18.24 km) 151, grid:east (19.553 km) 141.1, grid:west (33.267 km) 144.2, grid:north (33.994 km) 86.6
openweather	Wind speed (wind_speed)	grid:center (10.975 km) 1.908, grid:south (18.24 km) 2.141, grid:east (19.553 km) 2.362, grid:west (33.267 km) 1.383, grid:north (33.994 km) 1.701
purpleair	PM2.5 (pm25)	6752 (3.679 km) 4.759, 99797 (6.962 km) 6.45, 133994 (7.18 km) 0.07246, 293725 (8.647 km) 8.861, 97279 (8.764 km) 9.811, 166435 (8.846 km) 9.846, 293735 (9.495 km) 8.454, 222601 (10.473 km) 4.713, +63 more
tceq	Carbon monoxide (co)	482011035 (6.378 km) 0.1543, 482011039 (16.162 km) 0.1547, 482011052 (16.841 km) 0.3839
tceq	Nitrogen dioxide (no2)	482010416 (0.0 km) 7.306, 482011035 (6.378 km) 10.03, 482011039 (16.162 km) 5.498, 482011052 (16.841 km) 13.03, 482011066 (19.515 km) 8.732, 482010055 (19.781 km) 6.369, 480391004 (20.723 km) 4.094, 482010026 (20.843 km) 10.61, +4 more
tceq	Sulfur dioxide (so2)	482011035 (6.378 km) -0.3937, 482011039 (16.162 km) 0.03387, 482010051 (18.688 km) -0.4813

Ground observation locations in the stored 72-hour context
dashed rings are great-circle distance from the anomaly



Event-nearest wind vectors from stored values (arrow points downwind)



Claims

Mark exactly one box per claim: V (Valid), I (Invalid), or U (Unsure). To skip a claim, leave all three boxes blank; a blank is recorded as missing and never turned into Unsure. Each claim appears exactly once, and the number printed next to it is how I'll refer to that claim if I need to ask you about it.

Label definitions, from the labeling guide

Valid (V): the claim holds up at the precision it states, and the evidence in the packet supports it.

Invalid (I): the evidence in the packet contradicts the claim, the claim misstates a measurement, or the physical reasoning does not hold.

Unsure (U): you cannot tell from what is shown. This covers thin evidence, measurements that cannot resolve the claim, ambiguous wording, and anything outside what you are comfortable judging.

Missing or thin evidence means Unsure, not Invalid.

A cause that is plausible but unproven stays Unsure unless the evidence actually contradicts it.

Some claims bundle several statements into one sentence. Judge those by their parts. If every part earns the same verdict, use that verdict.

Claim 1 of 36

On July 15, 2026, at 21:00 UTC, nitrogen dioxide concentrations at the TCEQ monitor in Houston spiked to 18.8 ppb, representing a severe statistical anomaly relative to expected levels.

Mark exactly one:	☐ V	☐ I	☐ U
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Note (optional):

Claim 2 of 36

Regional photochemical accumulation is contradicted because openweather reported complete cloud cover near 21:00 UTC and openaq ozone remained very low at the anomaly location, conditions inconsistent with strong sunlight-driven ozone production as the main cause of the severe NO2 enhancement.

Mark exactly one:	☐ V	☐ I	☐ U
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Note (optional):

Claim 3 of 36

Meteorological trapping is supported because winds near the anomaly were very weak, with 0.0 m/s at the nearest asos station and 0.45 m/s in openweather at about 21:00 UTC, and because gfs indicated a shallow planetary boundary layer of about 114 m at 18:00 UTC, all of which would limit dilution of local NO2 emissions.

Mark exactly one:

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Note (optional):

Claim 4 of 36

Regional photochemical ozone-smog buildup is an unlikely primary cause of the NO2 spike because ozone near the anomaly was extremely low at the event time despite warm conditions, indicating the NO2 enhancement was dominated by fresh local emissions and titration instead of oxidized afternoon regional air.

Mark exactly one:

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Note (optional):

Claim 5 of 36

The openaq ozone sensor located 0.01 km from the anomaly site measured 0.002 ppm at 2026-07-15T21:00:00+00:00.

Mark exactly one:

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Note (optional):

Claim 6 of 36

Near-zero surface wind speeds recorded by ASOS and OpenWeather around 21:00 UTC on July 15, 2026, support horizontal atmospheric stagnation as a key factor in accumulating NO₂ detected by TCEQ.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 7 of 36

A fresh local combustion plume is supported because the anomalous tceq NO₂ concentration reached 18.8 ppb at 2026-07-15 21:00 UTC while co-located ozone was only 0.002 ppm at the same time, a combination consistent with recently emitted NO titrating ozone near a source rather than aged regional pollution.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 8 of 36

The PM_{2.5} levels are elevated at 5.4 ug/m³, which is higher than the mean of 7.2635 ug/m³.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 9 of 36

There is no evidence to suggest that other air pollutants, such as CO, SO₂, or O₃, contributed to the anomaly.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 10 of 36

At 20:55 UTC on July 15, 2026, ASOS meteorological observations near the anomaly reported calm surface winds of 0.0 m/s.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 11 of 36

A severe thunderstorm or tornado event occurred in the area, stirring up pollutants and reducing air quality.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 12 of 36

The air quality anomaly in Houston, Texas is caused by high levels of PM2.5.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 13 of 36

The tceq NO2 monitor at 29.686, -95.295 measured 18.8 ppb at 2026-07-15T21:00:00+00:00.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 14 of 36

GFS model estimates of a low boundary layer height near 113.7 meters near the anomaly location support reduced vertical atmospheric mixing during the high NO₂ event.

Mark exactly one:

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Note (optional):

Claim 15 of 36

Coincident near-surface ozone concentrations fell to 0.002 ppm at 21:00 UTC on July 15, 2026, indicating ozone titration by primary nitric oxide under suppressed photolysis conditions.

Mark exactly one:

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Note (optional):

Claim 16 of 36

OpenWeather reports of 100 percent cloud cover support diminished solar irradiance, which reduced photochemical photolysis of NO₂ and corresponds with low surface ozone values observed by OpenAQ.

Mark exactly one:

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Note (optional):

Claim 17 of 36

Surface meteorological conditions during the anomaly were characterized by calm wind speeds and dense cloud cover, restricting atmospheric dispersion.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 18 of 36

The air-quality chemistry at 2026-07-15 21:00 UTC is most consistent with a fresh local combustion plume because NO₂ was elevated while nearby ozone was only 0.002 ppm, indicating ozone titration by newly emitted NO_x rather than regional photochemical accumulation.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 19 of 36

High cloud cover and depressed ambient ozone levels suppressed the photochemical degradation of NO₂, facilitating its accumulation at the monitoring site.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 20 of 36

The anomaly is a result of a combination of factors including high temperatures, low humidity, and stagnant atmospheric conditions, leading to poor air mixing and increased pollutant concentrations.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 21 of 36

The NO₂ levels were also significantly higher than normal during the same period.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 22 of 36

The air quality anomaly in Houston, Texas is caused by low wind speeds.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 23 of 36

The anomaly occurred on July 15th at around 21:00 UTC, with the nearest data point being 0.0 min away.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 24 of 36

A fresh nearby combustion emission plume from mobile traffic, diesel equipment, rail, marine, or industrial combustion is a likely cause of the severe NO₂ spike because the monitor recorded 18.8 ppb NO₂ against an expected 6.0 ppb while co-located ozone was only 0.002 ppm, which is a classic near-source NO titration signature rather than aged regional pollution.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 25 of 36

Stagnant atmospheric conditions, characterized by near-zero wind speeds and a low planetary boundary layer height, restricted pollutant dispersion and trapped NO₂ near the ground.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 26 of 36

Meteorological conditions favored temporary trapping of local emissions because near-anomaly surface winds were extremely weak around the event and the modeled planetary boundary layer height was only about 114 m at 18:00 UTC.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 27 of 36

Weak dispersion and boundary-layer trapping are a likely cause of the elevated NO₂ concentration because winds near the anomaly were very light at about 0.0 to 0.45 m/s and the modeled planetary boundary layer height was low at about 114 m, conditions that favor accumulation of locally emitted NO_x near the surface.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 28 of 36

On July 15, 2026, at 21:00 UTC, TCEQ ground sensors recorded a severe NO₂ elevation reaching 18.8 ppb at coordinates (29.686, -95.295), corresponding to a z-score of 6.17 above the expected level of 6.01 ppb.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 29 of 36

At 21:00 UTC on July 15, 2026, openaq sensors co-located at the anomaly site recorded a surface ozone level of 0.002 ppm.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 30 of 36

The anomaly detection record for the tceq NO2 observation at 2026-07-15T21:00:00+00:00 reports an expected value of 6.012230215827339 ppb, a z-score of 6.171617557009788, and a severity classification of severe.

Mark exactly one:

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Note (optional):

Claim 31 of 36

The air quality anomaly is related to PM2.5 levels.

Mark exactly one:

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Note (optional):

Claim 32 of 36

The TCEQ monitor at 29.686, -95.295 measured a severe NO2 anomaly of 18.8 ppb at 2026-07-15 21:00 UTC, far above the expected 6.01 ppb baseline.

Mark exactly one:

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Note (optional):

Claim 33 of 36

The anomaly is caused by a nearby industrial facility releasing excessive amounts of particulate matter into the atmosphere.

Mark exactly one:

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Note (optional):

Claim 34 of 36

The air quality anomaly in Houston, Texas on July 15, 2026 was characterized by elevated levels of PM2.5.

Mark exactly one:

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Note (optional):

Claim 35 of 36

The air quality anomaly in Houston, Texas is caused by high levels of CO.

Mark exactly one:

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Note (optional):

Claim 36 of 36

Primary emissions from nearby industrial processes or urban vehicular traffic continuously supplied nitrogen oxides into the poorly mixed surface layer.

Mark exactly one:

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Note (optional):

Optional: most likely cause

Your best guess at the true cause of this event, in your own words. "Insufficient evidence to determine a single cause" is a perfectly fine answer.
